

```
# Install required libraries
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.0)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2021.8.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.0)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.2)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.4)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.9.1)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.4.3)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.6)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.2)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.3.0)
Requirement already satisfied: numpy<=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.19.0)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.13.0)
Requirement already satisfied: pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.10.13)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (59.0.0)
Requirement already satisfied: packaging<=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (20.0)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.41.4)
Requirement already satisfied: typing-extensions<=4.14.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.14.1)
Requirement already satisfied: typing-inspection<=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.3.0)
Requirement already satisfied: certifi<=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2017.4.17)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.0.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.7.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (5.2.1)
Requirement already satisfied: MarkupSafe<=2.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open) (1.15.0)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-3.8.0/en\_core\_web\_sm-3.8.0.tar.gz 12.8/12.8 MB 92.2 MB/s eta 0:00:00
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
# Import libraries
import nltk
import spacy
from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer
```

```
# Download NLTK resources
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
True
```

```
medical_text = """
Diabetes is a chronic disease that affects how the body processes blood sugar.
If untreated, diabetes may cause heart disease, kidney failure, nerve damage and vision problems.
Early diagnosis and proper treatment help improve patient outcomes.
"""
```

```
import nltk
nltk.download('punkt_tab') # Download the missing resource
sentences = sent_tokenize(medical_text)
for i, sentence in enumerate(sentences, 1):
    print(f"Sentence {i}: {sentence}")
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
Sentence 1:
Diabetes is a chronic disease that affects how the body processes blood sugar.
Sentence 2: If untreated, diabetes may cause heart disease, kidney failure, nerve damage and vision problems.
Sentence 3: Early diagnosis and proper treatment help improve patient outcomes.
```

```
words_nltk = word_tokenize(medical_text)
print(words_nltk)
```

```
['Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the', 'body', 'processes', 'blood', 'sugar', 'if', 'untreated', 'diabetes', 'may', 'cause', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'and', 'vision', 'problems.', 'early', 'diagnosis', 'and', 'proper', 'treatment', 'help', 'improve', 'patient', 'outcomes.']
```

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(medical_text)

words_spacy = [token.text for token in doc]
print(words_spacy)
```

```
['\n', 'Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the', 'body', 'processes', 'blood', 'sugar', 'if', 'untreated', 'diabetes', 'may', 'cause', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'and', 'vision', 'problems.', 'early', 'diagnosis', 'and', 'proper', 'treatment', 'help', 'improve', 'patient', 'outcomes.']
```

```
stemmer = PorterStemmer()

stemmed_words = [stemmer.stem(word) for word in words_nltk if word.isalpha()]
print(stemmed_words)
```

```
['diabet', 'is', 'a', 'chronic', 'diseas', 'that', 'affect', 'how', 'the', 'bodi', 'process', 'blood', 'sugar', 'if', 'untreated', 'diabetes', 'may', 'cause', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'and', 'vision', 'problems.', 'early', 'diagnosis', 'and', 'proper', 'treatment', 'help', 'improve', 'patient', 'outcomes']
```

```
lemmatizer = WordNetLemmatizer()

lemmatized_words_nltk = [lemmatizer.lemmatize(word.lower())
                          for word in words_nltk if word.isalpha()]
print(lemmatized_words_nltk)
```

```
['diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar', 'if', 'untreated', 'diabetes', 'may', 'cause', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'and', 'vision', 'problems.', 'early', 'diagnosis', 'and', 'proper', 'treatment', 'help', 'improve', 'patient', 'outcomes']
```

```
lemmatized_words_spacy = [token.lemma_ for token in doc if token.is_alpha]
print(lemmatized_words_spacy)
```

```
['Diabetes', 'be', 'a', 'chronic', 'disease', 'that', 'affect', 'how', 'the', 'body', 'process', 'blood', 'sugar', 'if', 'untreated', 'diabetes', 'may', 'cause', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'and', 'vision', 'problems.', 'early', 'diagnosis', 'and', 'proper', 'treatment', 'help', 'improve', 'patient', 'outcomes']
```

```

import nltk
import spacy
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

# Downloads (run once)
nltk.download('punkt')

# Load spaCy model
nlp = spacy.load("en_core_web_sm")

# Medical text
text = """
Diabetes is a chronic disease that affects how the body processes blood sugar.
If untreated, diabetes may cause heart disease, kidney failure, nerve damage and vision problems.
Early diagnosis and proper treatment help improve patient outcomes.
"""

# Tokenization
tokens = word_tokenize(text)
original_words = [word for word in tokens if word.isalpha()]

# Stemming
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(word) for word in original_words]

# Lemmatization (spaCy)
doc = nlp(text)
lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]

# Display output together
print(f"{'ORIGINAL':<15}{'STEMMING':<15}{'LEMMATIZATION'}")
print("-" * 45)

for o, s, l in zip(original_words, stemmed_words, lemmatized_words):
    print(f"{o:<15}{s:<15}{l}")

```

[nltk_data] Downloading package punkt to /root/nltk_data...

[nltk_data] Package punkt is already up-to-date!

ORIGINAL	STEMMING	LEMMATIZATION
Diabetes	diabet	Diabetes
is	is	be
a	a	a
chronic	chronic	chronic
disease	diseas	disease
that	that	that
affects	affect	affect
how	how	how
the	the	the
body	bodi	body
processes	process	process
blood	blood	blood
sugar	sugar	sugar
If	if	if
untreated	untreat	untreate
diabetes	diabet	diabete
may	may	may
cause	caus	cause
heart	heart	heart
disease	diseas	disease
kidney	kidney	kidney
failure	failur	failure
nerve	nerv	nerve
damage	damag	damage
and	and	and
vision	vision	vision

problems	problem	problem
Early	earli	early
diagnosis	diagnosi	diagnosis
and	and	and
proper	proper	proper
treatment	treatment	treatment
help	help	help
improve	improv	improve
patient	patient	patient
outcomes	outcom	outcome

Start coding or [generate](#) with AI.